

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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2.1 Job Profile Data

The first dataset consists of all publicly available professional LinkedIn profiles between January 2021 through June 2025. The dataset includes approximately 75 million employment spells from 23 million unique individuals across 2.5 million UK-based firms. Each observation includes the worker’s job title, employer, start and end dates (if available), seniority level,³ and compensation.⁴ The data provide near-real-time employment stocks, which enable us to track within-firm workforce composition changes over time.

From the panel of individual LinkedIn profiles, we construct a monthly firm-level panel from January 2021 through June 2025. For each firm-month, we calculate total employment as the count of unique individuals with active employment spells at a certain company in that month, and log-transform this variable. This is our main employment stock measure. We then separate total employment into junior and senior employment stocks based on Revelio seniority classifications of jobs. We also calculate average compensation as the log of the mean of reported salaries plus bonuses in a firm. We only focus on firms that remained in business throughout the sample period.

To calculate firm-level exposure scores, we use the assessment framework developed by Eloundou et al. (2024). They evaluate 19,265 detailed work activities and tasks from the O*NET 27.2 database to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. We focus on their ‘E1’ measure, which captures exposure to current LLMs, rather than LLMs combined with additional software. We do so because it is the most appropriate metric for identifying short-term labor market effects that do not require additional software development or complex organizational changes. To obtain our firm-level

³Revelio Labs determines seniority using an ensemble model that combines three components: (i) current job characteristics including title, company, and industry; (ii) employment history including tenure and previous position seniority; and (iii) individual age. These three scores are averaged to create a continuous seniority metric, which is then mapped to seven ordinal levels based on keyword benchmarks (e.g., “junior,” “senior,” “director”). The levels range from Entry Level (e.g., interns, trainees) through Junior, Associate, Manager, and Director levels, to Executive and Senior Executive levels (e.g., C-suite positions).

⁴Revelio Labs estimates compensation using a predictive model trained on publicly available visa application data, self-reported salaries, and job posting information. The model incorporates position-specific variables (job title, seniority level, company, location), individual-specific factors (tenure at company), and temporal controls (year of observation) to generate salary predictions for positions where compensation data are not directly observed.

ket characteristics. We examine heterogeneity along two dimensions: compensation levels (splitting firms and occupations by median pre-treatment salary) and geographic concentration (comparing London-based versus non-London entities). These dimensions are highly policy-relevant, as compensation-level heterogeneity provides a first indication as to whether LLMs may compress or widen wage inequality, while geographic variation tests whether disruption concentrates in London’s knowledge economy or extends throughout the UK labor market.

4.3.1 Heterogeneity by Compensation Level

Table A2 examines how firm-level employment effects vary by average compensation, with high-compensation jobs serving as a proxy for more highly educated white collar work, and low-compensation jobs for relatively low-educated blue collar work. We split firms at the median pre-treatment compensation level. Panel A shows that high-compensation firms experience substantially larger employment reductions than their lower-paying counterparts. A one standard deviation increase in LLM exposure reduces total employment at high-compensation firms by 0.7 percent ($p = 0.009$), driven primarily by a 0.8 percent reduction in low-seniority positions ($p = 0.005$). In contrast, Panel B indicates that low-compensation firms show no significant employment effects, with point estimates near zero for both total employment and low-seniority positions.

This pattern extends to hiring intentions. Table A4 shows that a one standard deviation increase in LLM exposure reduces the probability of posting vacancies by 1.3 percentage points at high-compensation firms ($p < 0.001$), compared to a 0.7 percentage point reduction at low-compensation firms ($p = 0.001$). While both groups curtail hiring, the effect is approximately twice as large at high-compensation firms.

The occupation-level analysis in Table A5 provides further evidence that the employment effects are concentrated among the highly paid. In high-salary occupations, a one standard deviation increase in LLM exposure is associated with a 5.7 percent reduction in job postings ($p = 0.009$) and £782 salary declines ($p < 0.001$), while low-salary occupations show no significant changes in posting volumes and minimal salary effects.

These results suggest that LLM disruption is almost entirely driven by changes

A1 Additional Tables

Table A1: Effect of LLM exposure on firm-level job listings

	Job postings
LLM exposure x Post	−0.010*** (0.002)
Firm fixed-effects	Yes
Industry-by-period fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.522

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity. The dependent variable is an indicator equal to one if firm i posted at least one job vacancy in month t . LLM exposure is the firm’s standardized pre-treatment exposure score, calculated as the employment-weighted average of occupational LLM exposure scores from Eloundou et al. (2024) using the workforce composition on 30 October 2022. All regressions include firm fixed effects and industry-by-period fixed effects. All other definitions are as in Table 3.

A2 Heterogeneous Treatment Effects

A2.1 Firm-Level Employment Stocks

Table A2: Effect of LLM exposure on firm-level job outcomes, high vs. low compensation firms

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
Panel A: High compensation firms					
LLM exposure x Post	−0.007*** (0.003)	−0.008*** (0.003)	−0.005* (0.003)	−0.001* (0.0004)	195.240* (101.538)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	634,030	633,977	634,018	633,977	634,030
Adj. R-squared	0.975	0.973	0.974	0.963	0.956
Panel B: Low compensation firms					
LLM exposure x Post	0.0001 (0.003)	−0.001 (0.003)	−0.00002 (0.003)	−0.0004 (0.0003)	69.929 (46.938)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	633,983	633,978	632,313	633,978	633,983
Adj. R-squared	0.988	0.988	0.985	0.956	0.946

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for high- and low-compensation firms. Firms are split at the median pre-treatment average compensation level (October 2022). Panel A presents results for firms above the median compensation, Panel B for firms below the median. All other definitions are as in Table 3.

Table A3: Effect of LLM exposure on firm-level job outcomes, London vs. Non-London firms

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
Panel A: HQ in London					
LLM exposure x Post	-0.001 (0.004)	-0.002 (0.004)	-0.004 (0.004)	-0.001 (0.001)	214.807 (132.316)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	377,175	377,129	376,920	377,129	377,175
Adj. R-squared	0.982	0.981	0.981	0.978	0.975
Panel B: HQ outside London					
LLM exposure x Post	-0.004** (0.002)	-0.005** (0.002)	-0.002 (0.003)	-0.001** (0.0003)	71.962 (64.222)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	890,838	890,826	889,411	890,826	890,838
Adj. R-squared	0.983	0.983	0.981	0.979	0.979

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for firms headquartered in London, and those headquartered elsewhere. Panel A presents results for London-based firms, Panel B for firms based outside London. All other definitions are as in Table 3.

A2.2 Firm-Level Job Postings

Table A4: Effect of LLM exposure on firm-level job listings, heterogeneity

	High compensation firms	Low compensation firms	HQ in London	HQ outside London
LLM exposure x Post	-0.013*** (0.003)	-0.007*** (0.002)	-0.016*** (0.003)	-0.008*** (0.002)
Firm fixed-effects	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes
Observations	577,489	690,524	377,175	890,838
Adjusted R ²	0.486	0.554	0.517	0.525

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity across different firm characteristics. Columns 1-2 split firms by median pre-treatment average compensation level. Columns 3-4 split firms by headquarter location (London versus outside London). All other definitions are as in Table A1.

Table A6: Effect of LLM exposure on occupation-level job outcomes, London vs. non-London

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: Job in London				
LLM exposure x Post	-0.034** (0.017)	-562.818*** (139.355)	-0.010*** (0.002)	74.609 (132.525)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,734	16,734	16,734	16,625
Adjusted R ²	0.954	0.922	0.926	0.605
Panel B: Job outside London				
LLM exposure x Post	-0.040** (0.018)	-475.947*** (141.444)	-0.011*** (0.003)	-119.267 (111.131)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,812	16,812	16,812	16,799
Adjusted R ²	0.961	0.938	0.945	0.683

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for London-based jobs and jobs based outside London. Panel A presents results for job openings in London, Panel B for job openings outside London. All other definitions are as in Table 4.

A2.3 Occupation-Level Job Listings

Table A5: Effect of LLM exposure on occupation-level job outcomes, high vs. low salary occupations

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: High salary occupations				
LLM exposure x Post	-0.057*** (0.022)	-782.371*** (184.426)	-0.013*** (0.003)	-354.844*** (131.069)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,413	8,413	8,413	8,412
Adjusted R ²	0.964	0.903	0.901	0.726
Panel B: Low salary occupations				
LLM exposure x Post	-0.020 (0.031)	-112.614 (179.137)	-0.004 (0.004)	240.391 (170.804)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,402	8,402	8,402	8,392
Adjusted R ²	0.961	0.817	0.860	0.436

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for high- and low-salary occupations. Occupations are split at the median pre-treatment average posted salary (October 2022). Panel A presents results for occupations above the median salary, Panel B for occupations below the median. All other definitions are as in Table 4.